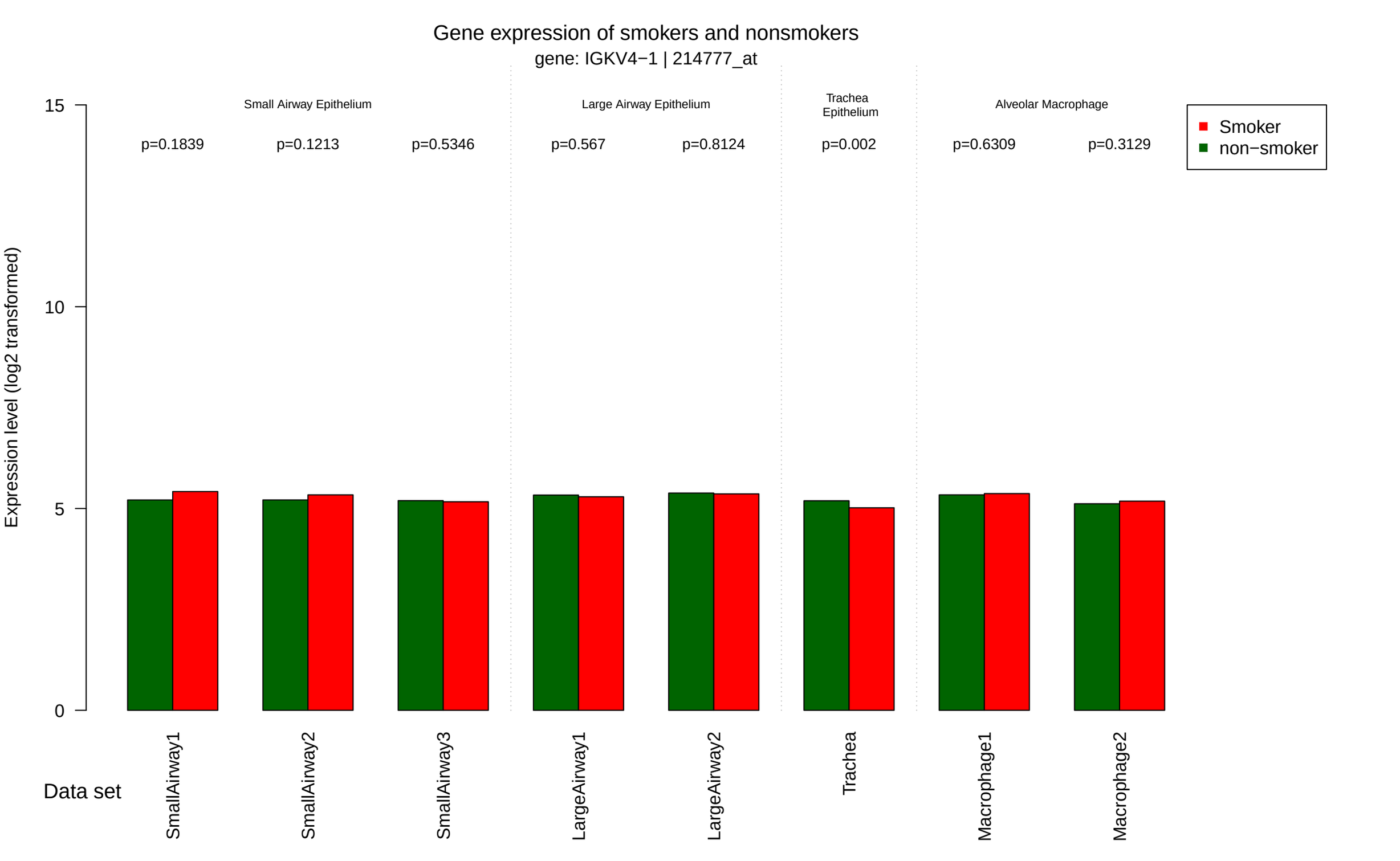
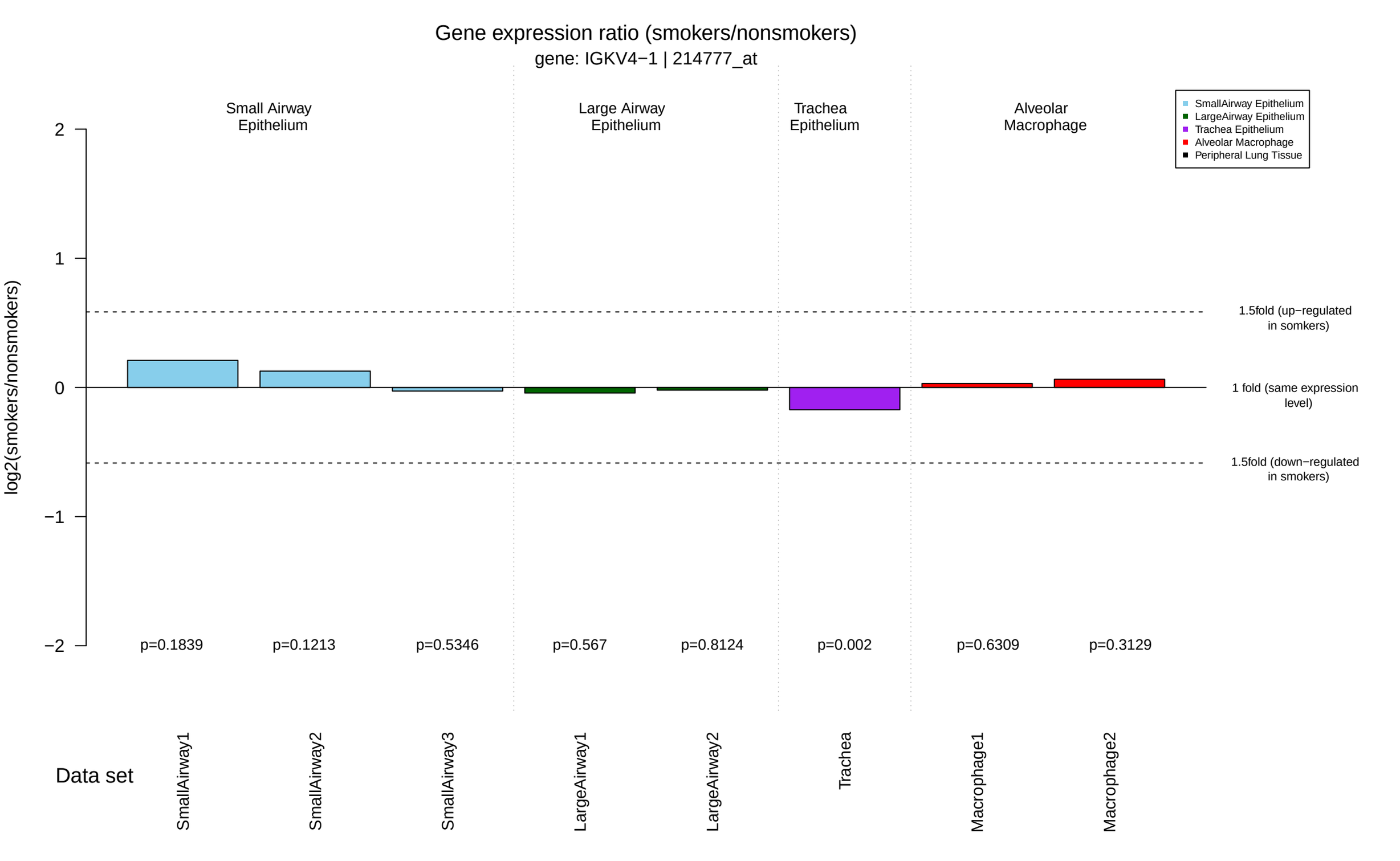
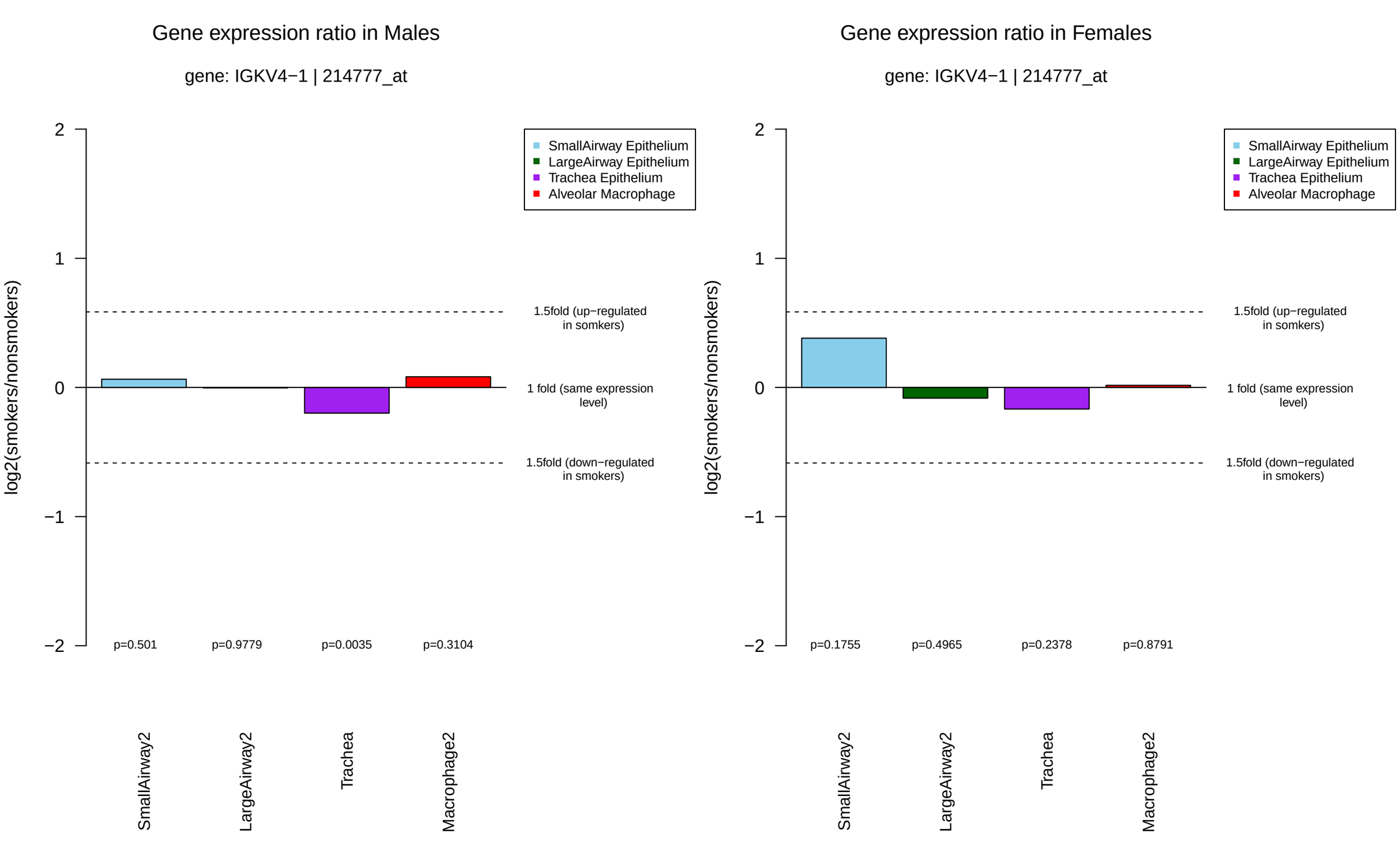
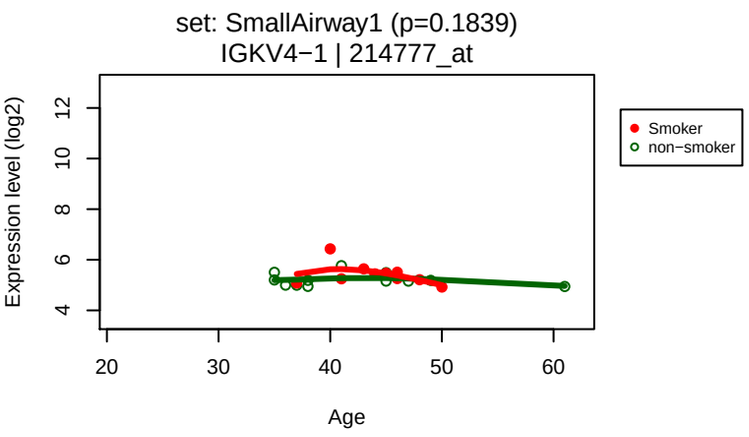
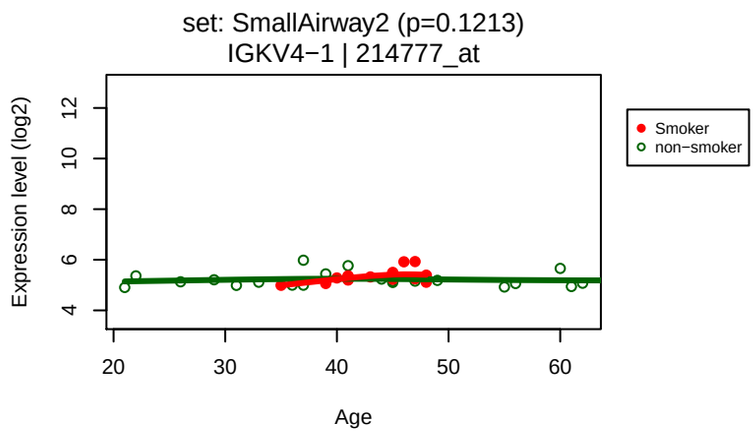
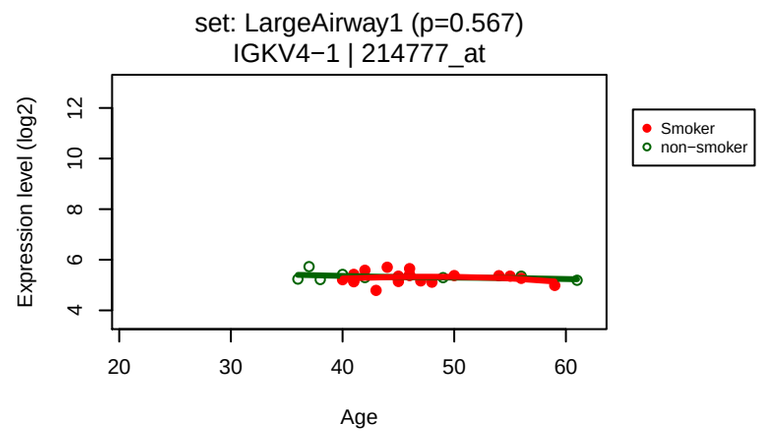

Gene expression of smokers and nonsmokers
gene: IGKV4−1 | 214777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1839
p=0.1213
p=0.5346
p=0.567
p=0.8124
p=0.002
p=0.6309
p=0.3129
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

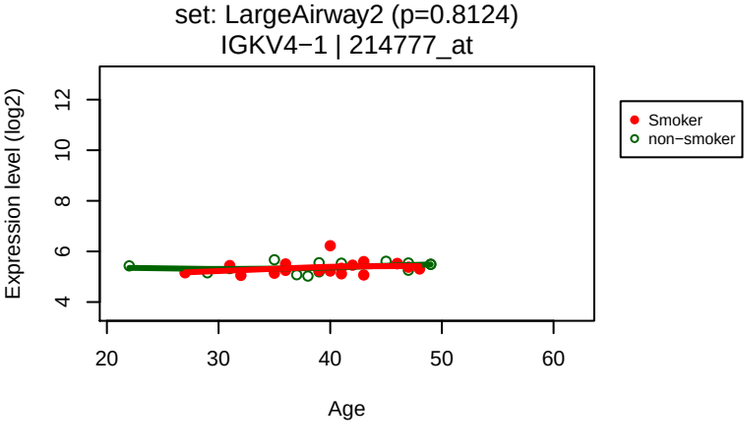

Gene expression ratio (smokers/nonsmokers)
gene: IGKV4−1 | 214777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1839
p=0.1213
p=0.5346
p=0.567
p=0.8124
p=0.002
p=0.6309
p=0.3129
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

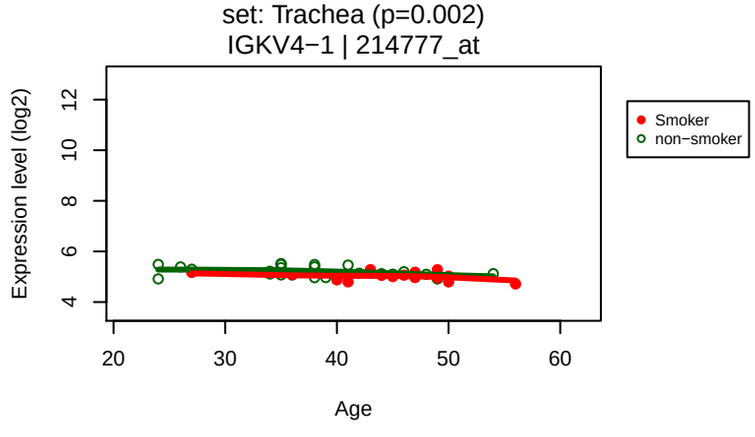

Gene expression ratio in Males
gene: IGKV4−1 | 214777_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.501
p=0.9779
p=0.0035
p=0.3104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGKV4−1 | 214777_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1755
p=0.4965
p=0.2378
p=0.8791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

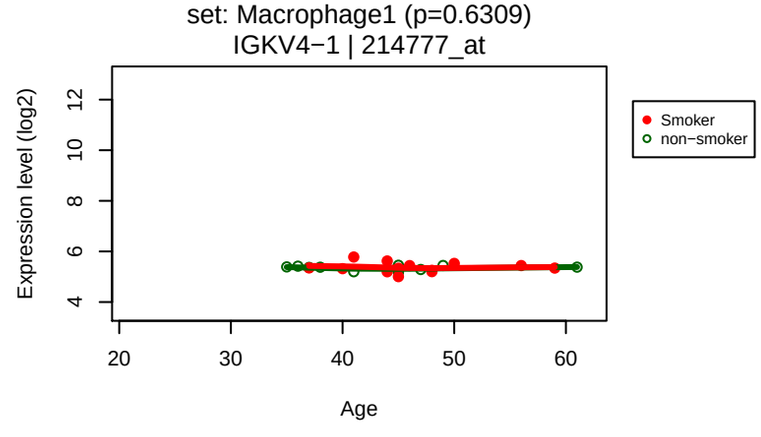

set: SmallAirway1 (p=0.1839)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker

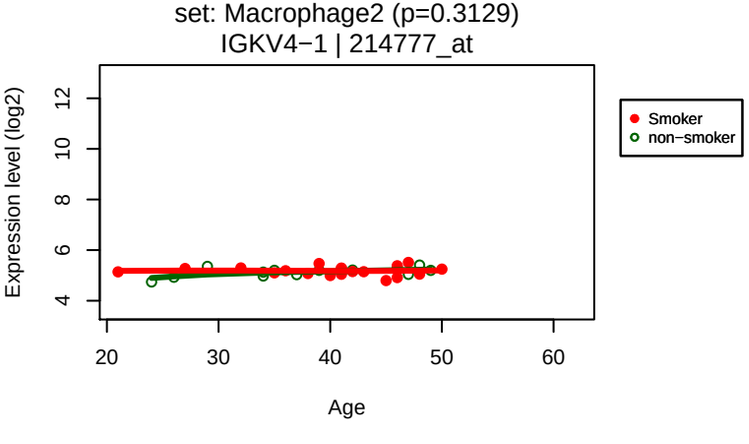

set: SmallAirway2 (p=0.1213)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker

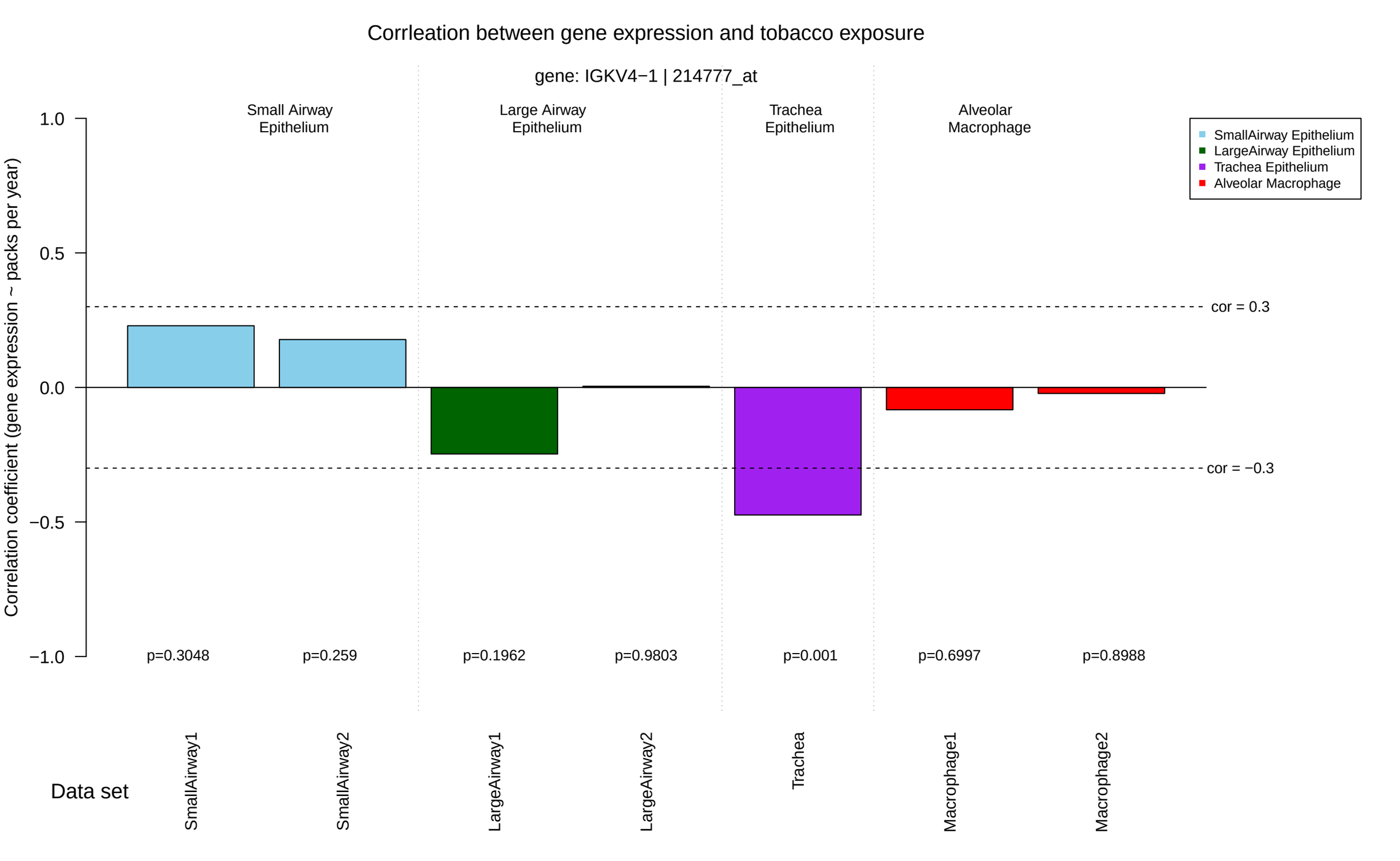

set: LargeAirway1 (p=0.567)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8124)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.002)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6309)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3129)
IGKV4−1 | 214777_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGKV4−1 | 214777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3048
p=0.259
p=0.1962
p=0.9803
p=0.001
p=0.6997
p=0.8988
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2